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Soil-erosion events on arable land are nowcast by machine learning

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ABSTRACT

Accurate estimates of the location, timing, and severity of soil-erosion events on arable land have eluded erosion-prediction technology for decades. Here, for the first time, we demonstrate how a machine learning model parameterised with spatiotemporal covariates within a back-end infrastructure of data cubes can nowcast the occurrence and relatively rank the severity of erosion events on arable field parcels at the regional scale with high accuracy and interpretable outputs. Our findings pave the way for dynamic erosion-monitoring systems to achieve healthy soils and improve food security.

1. Introduction

Soil erosion on arable land is a major and unresolved environmental threat to soil and food security, worldwide (Quinton and Fiener, 2024). In the short term, extreme erosion events lead to crop damages and harvest losses (Batista et al., 2023) that can promptly affect food supply chains; in the long term, erosion leads to a deterioration of soil health that may compromise future crop yields and food production (Quinton et al., 2022). Both short- and long-term erosional threats to food systems are exacerbated by climate-change-driven increases in the amount, intensity, and frequency of erosive rainfall (Auerswald et al., 2024). However, developing effective erosion-mitigation strategies at policy-relevant scales (e.g., regional, national, or continental) is challenging due to the highly episodic and localised nature of soil-erosion events. That is, erosion does not happen with the same intensity and on the same fields for every cropping season (Evans et al., 2016; Fiener et al., 2019), as erosion drivers are subject to a large spatiotemporal variability (Möller et al., 2017) and some degree of randomness effected by farmer decisions (Auerswald et al., 2018).

Current tools for regional-scale erosion-risk assessments, such as those employed at the EU (Borrelli et al., 2022), rely almost entirely on empirical models originally developed in the USA for estimating long-term average soil losses from uniform slope segments. Such approaches are unable to represent the spatiotemporal variability of erosion processes in complex agricultural landscapes and have been subject to severe criticism (Evans and Boardman, 2016; Parsons, 2019). Process-oriented models provide event-based, quantitative erosion simulations, though their application is typically confined to small watersheds where model parameters can be measured and/or calibrated (Brunner et al., 2023). Similarly, field-based methods are restricted by the adequacy of sampling (Parsons, 2019), as erosion cannot be measured everywhere all the time. Hence, there is a lack of reliable tools for continuously estimating when, where, and how severely are arable fields affected by erosion events (Parsons, 2019). This situation restricts the development of large-scale erosion-monitoring systems to promote soil health and food security as envisioned in the new European soil monitoring law (Panagos et al., 2024).

Here, we propose a new approach for addressing the erosion detection problem, which capitalises on well-established machine learning methods, recent developments in big geodata infrastructures (Baumann,

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2021), and spatiotemporally contiguous rain-radar measurements (Winterrath et al., 2012) to provide a dynamic basis for erosion mapping. Specifically, we demonstrate how a random-forest model can detect the occurrence and relatively rank the severity of erosion events on arable field parcels at the regional scale with high accuracy and interpretable outputs. This is achieved using multi-dimensional spatiotemporal covariates within a back-end infrastructure of data cubes and server-based computing capacities and training data from multiple erosion events across 1790 arable field parcels in Bavaria, Germany (Fig. 1).

2. Observational training data

The erosion training data were acquired by visually interpreting the presence of erosion features on aerial images taken shortly after erosive rainfall events during spring/summer 2011 and 2012 (Auerswald et al., 2018; Fischer et al., 2018). An erosion event on a field parcel was detected if erosion features, i.e. rills, ephemeral gullies, or sediment accumulation on neighbouring areas were identified from the aerial images. Erosion severity within each analysed field parcel was ranked into four classes according to the proportion of the area with visible signs of erosion (Fischer et al., 2018): Class 0 = no signs of soil erosion; Class 1 = < 10 % of the field area showed erosion features clear enough to be seen in the aerial image; Class 2 = 10 % – 30 % of the field area showed erosion features; and Class 3 = > 30 % of the field area showed erosion features. Interrill areas supplying sediment to rills, gullies, or neighbouring fields were also considered in the estimation of the erosion-affected area. However, in the absence of such features, interrill erosion was not detected.

The images were inspected by four experts for quality control, and the median of the expert classifications was used as observational data for model development (Supplementary Information Fig. 1). Fields in which the resulting median was non-integral were removed from further analysis to preserve the original number of visual erosion classes and to avoid including data with a lack of majority agreement among the classifiers. A shapefile from the Bavarian land parcel identification system was used to recognise fields during the image analysis and to filter out non-arable land uses. Ultimately, 1790 individual arable field parcels with an erosion classification following 18 different rainfall events were used for model training and testing (Supplementary Information Fig. 4).

3. Spatiotemporal covariates

Five sets of spatiotemporal covariates were considered: (i) rainfall indices derived from hourly ground-truth-adjusted radar measurements (1 km, daily resolution) (Dierks and Möller, 2024; Winterrath et al., 2012) (ii) interpolated phenological stages for the main crop types in Bavaria (1 km, daily resolution) (Möller, 2021; Möller et al., 2020); (iii) a gap-filled Landsat-like NDVI time series produced with the Enhanced Spatial and Temporal Adaptive Reflectance Fusion Model (ESTARFM) to blend MODIS and Landsat (7 and 8) imagery (30 m, weekly resolution) (Zhu et al., 2010); (iv) parcel-specific information on crop type, crop management, and field sizes taken from the Bavarian land parcel

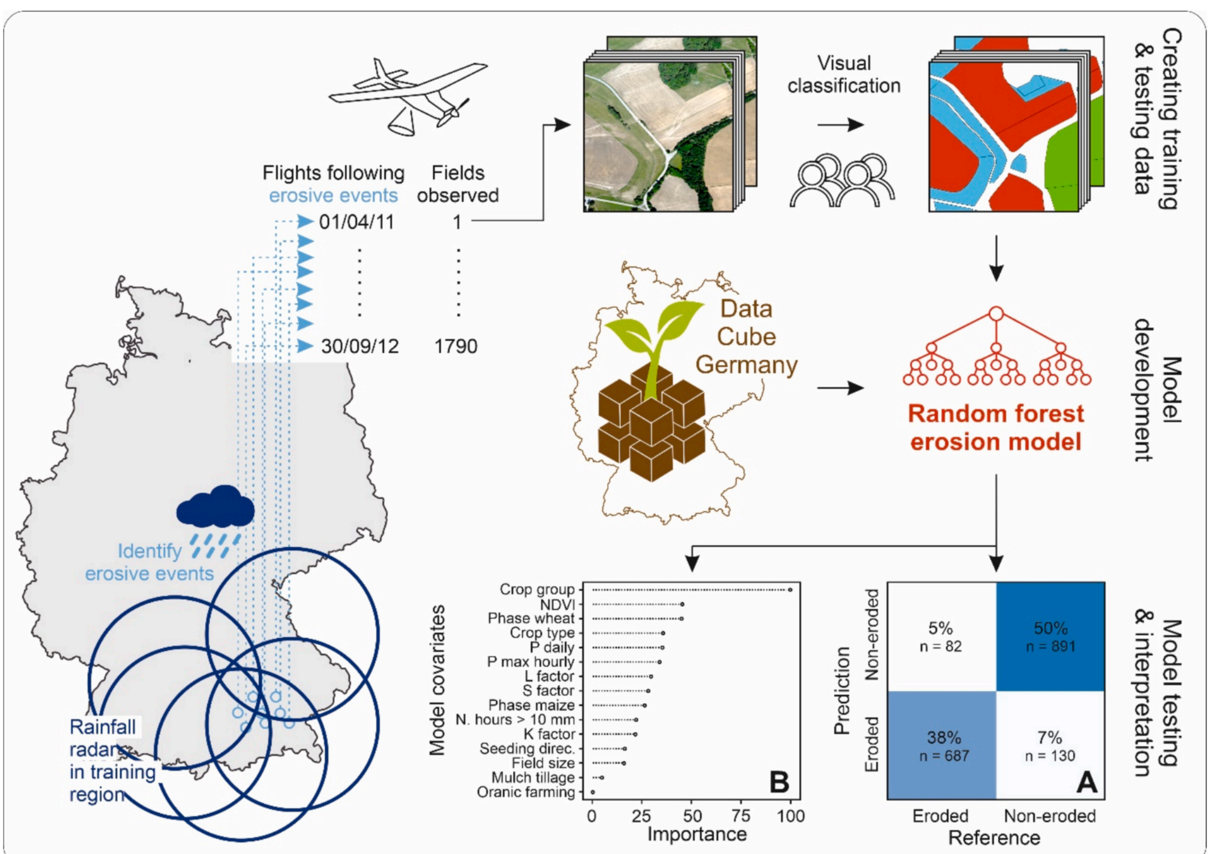

Fig. 1. Schematic overview of the regional machine learning erosion model. Visually classified aerial images and multidimensional spatiotemporal covariates are used to train and test a random forest. A data cube from the Julius Kühn Institute (JKI) (Beyer et al., 2023) provides Germany-wide web-coverage services for the dynamic parameterisation of the model. Panels display the variable importance ranking (A) and confusion matrix (B) for the model trained to detect the occurrence of erosion events on arable land. Legend: K, L, and S factors represent soil erodibility, slope length, and slope gradient effects in the Universal Soil Loss Equation, respectively; NDVI is the normalised difference vegetation index; P is precipitation; N. hours > 10 mm is the number of hours per day with precipitation above 10 mm h^{-1}; Phase is crop phenological phase. Detailed information on the model variables is provided in section 3.

identification system (vector based, annual resolution); and v) soil erodibility, slope gradient, and slope length parameters extracted from a regional erosion model (5 m resolution, temporally static) (Fischer et al., 2018) (Supplementary Information Table 1). Median values of covariates per field parcel were calculated as input data for the modelling. In the case of temporally dynamic covariates, we considered the closest date to the erosion event correspondent to the aerial image used to derive the erosion classification for each field. Hence, model estimates are made at the field parcel level and with a daily temporal resolution.

4. Model development and evaluation

For modelling, we employed a random forest algorithm (Breiman, 2001) widely used for spatiotemporal predictions (Hengl et al., 2018) that makes estimates through an ensemble of decision trees based on bootstrap samples. Two target variables were evaluated: (i) the occurrence of erosion events per field parcel (i.e. detection of classes > 0) and (ii) the severity of erosion events on field parcels (i.e., ranking classes 1 – 3). Moreover, the model was trained and tested with the complete dataset comprising 68 different types of arable crops and individually for the main crop groups in Bavaria (i.e. maize and winter cereals, which cover 44 % and 34 % of arable land in Bavaria, respectively). In all cases, model performance was assessed with a 4-fold cross-validation. To prevent overfitting, we did not use date/time information as covariates (Meyer et al., 2018). Moreover, we balanced the training data for the erosion-ranking- and crop-group-targeted models by majority under sampling, which was repeated 30 times and showed stable results above 20 iterations. This approach avoids over-optimistic model-testing metrics that are biased towards the majority class in unbalanced datasets (Taghizadeh-Mehrjardi et al., 2020). Data analysis and model development were performed in R (v. 4.1.1) (R Core Team, 2021) using the *randomForest* (v. 4.7–1) (Liaw and Wiener, 2002) and *caret* (v.6.0–89) (Kuhn, 2008) packages. Random forest hyperparameters were not tuned and *mtry* was set in each model as the square root of the number of covariates.

5. Detecting and ranking the severity of soil-erosion events

Our approach can detect erosion events on arable parcels with an overall accuracy of 88 % (recall = 89 %, precision = 84 %) (Fig. 1A). Land-management covariates (crop group, crop type, phenological phase, and NDVI), combined with rainfall indices (daily total precipitation and daily maximum hourly precipitation intensity), are the most powerful predictors, according to a feature importance ranking of the model (Fig. 1B). This is congruent with the basic notion that erosion on arable land typically occurs when erosive rainfall strikes unprotected soil and that certain crops are particularly erosion-prone due to interactions between crop phenology, crop management, and weather. For instance, conventional maize is highly erosion-prone in Bavaria due to late sowing dates that lead to a time window of low canopy cover that coincides with the rise of erosive rainfall events in late spring (Auerswald and Menzel, 2021).

Within the eroding field parcels, the model ranks erosion events into severity classes with an overall accuracy of 50 % (Fig. 2A), which is within the uncertainty levels in the observational training data (Supplementary Information Fig. 1). It is worth highlighting that, similar to the visual classification, errors are predominantly associated with neighbouring severity classes and that the most severe erosion class, arguably the main prediction target, is detected with higher accuracy (recall = 63 %, precision = 55 %) (Fig. 2A). A slope-gradient-effect covariate was the most important feature influencing the model's capacity to rank the severity of the erosion events per field parcel (Fig. 2B). This is explained by the fact that extreme erosion events tend to only occur in sloped, sparsely vegetated land during erosive rainfall, while flatter fields will suffer less erosion, or will not erode at all, under equivalent conditions (Evans et al., 2016; Prasuhn, 2020).

As the model results display a large sensitivity to crop group and considering that our training data is biased towards eroding maize fields (Supplementary Information Fig. 2), we tested our approach in isolation for the two main crop groups in our dataset, i.e. maize and winter cereals. The occurrence of erosion events on maize fields is detected with an overall accuracy of 74 % (recall = 76 %, precision = 73 %) (Supplementary Information Fig. 3). In this case, the model displays a higher sensitivity to rainfall and topographical covariates, highlighting how erosion can occur on row crops under a relatively wide range of soil cover conditions (Supplementary Information Fig. 4) if stream power is sufficiently high to promote rill initiation. The occurrence of erosion events on winter-cereal fields is detected with an overall accuracy of 84 % (recall = 80 %, precision = 86 %), almost exclusively during time windows of low canopy cover (Supplementary Information Figs. 3 and 4). This is because winter cereals have lower plant heights and narrower row spacing than maize. Such characteristics lead to a lower fall height of intercepted rainfall and provide a fine-grained pattern of ground cover, both of which decrease erosion propensity in periods of greater vegetation strength (Supplementary Information Fig. 4).

The main limitations of our modelling approach stem from the

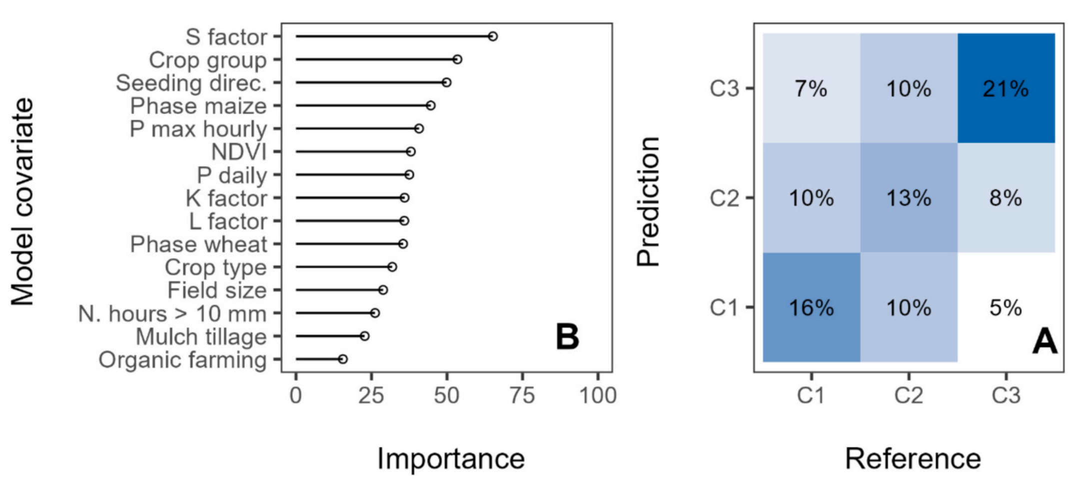

Fig. 2. Variable importance ranking (A) and confusion matrix (B) for the random-forest model trained for ranking the erosion-severity classes of the eroding arable fields (n = 552). Legend: K, L, and S factors represent soil erodibility, slope length, and slope gradient effects in the Universal Soil Loss Equation, respectively; NDVI is the normalised difference vegetation index; P is precipitation; N. hours > 10 mm is the number of hours per day with precipitation above 10 mm h^{-1}; Phase is crop phenological phase. Detailed information on the model variables is provided in the methods section.

observational data. The subjectivity in the visual classification of erosion features adds uncertainty to the model, which cannot be expected to be more precise than the measurements in the training data. In addition, we cannot identify interrill erosion from the aerial images in the absence of linear erosion features or sediment accumulation on neighbouring fields, which means we are likely to underestimate the detection of low-magnitude erosion events. Moreover, the observational data are biased towards spring/summer heavy rainfall events and eroding maize fields (Supplementary Information Figs. 2 and 5). This currently leads to a restriction in model applicability in terms of seasons and crop types. However, it should be noticed that our observational data cover the most erosion-prone crops, soils, slopes, slope lengths, and time windows in Bavaria (Supplementary Information Fig. 6). This is particularly important because soil erosion is an episodic phenomenon; therefore, the observational erosion data must be specifically representative of the conditions that trigger erosion events. Arguably, it is much easier to acquire further observational data for situations where erosion does not occur, which could contribute towards a wider applicability of our approach in, e.g. Bavaria. Nevertheless, for wider applications of our modelling framework, we recommend defining areas of applicability based on the characteristics of the training data and the predictor space, as suggested in Meyer and Pebesma (2021), to avoid spurious extrapolations of data-driven models.

6. The way forward: Dynamic erosion monitoring systems to improve soil health and food security

The main breakthrough achieved by our modelling approach is that, for the first time, soil-erosion events on individual field parcels can be nowcast. This is enabled by the use of dynamic covariates that can be immediately retrieved at any time from the JKI data cube or from the German Weather Service, which provides clients with precipitation nowcasts for a lead time of up to two hours (Winterrath et al., 2012). Hence, we created a framework for monitoring soil-erosion events across potentially large areas, in close-to-real time, and with a high spatial resolution – all of which will be paramount for identifying (i) crop damage and potential harvest losses associated with extreme erosion events and (ii) fields that are frequently exposed to erosion and are therefore likely to suffer from medium- to long-term erosional effects on soil health.

Moreover, model outputs have a spatiotemporal resolution that is relatable to farmers and other end users, which can increase the involvement of stakeholders in citizen-science projects and, in turn, increase model reliability (Blair et al., 2019). Importantly, as repeated observations from the same field parcels become available, the algorithm can learn about field peculiarities that would otherwise not be represented in the parameterisation of large-scale models. This is specifically relevant if infield erosion is repeatedly initiated due to upslope inflow from linear landscape features such as small ditches along the street network (Prasuhn, 2020). The focus on erosion events will also allow for a better assessment of offsite damages to infrastructure and private property caused by episodic, extreme events in upland fields, which are diluted in long-term, averaged-out risk assessments.

The outputs from our modelling framework are simple enough to be understood by non-experts and falsified by easily measured field data. In addition, outputs are always accompanied by a forward uncertainty estimation that helps to inform decision-making, as random forests are ensemble-based classification algorithms (Breiman, 2001). There are also different adaptations of random forests and other machine learning algorithms that can be used to represent model uncertainty through probabilistic predictions (Schmidinger and Heuvelink, 2023). Another beneficial aspect of machine learning models is the focus on observational data and model testing – both of which have been recently missing from large-scale erosion modelling (Batista et al., 2019).

Ultimately, the ability of our modelling framework to detect erosion events with a high accuracy based on interpretable covariates that are compatible with our understanding of soil-erosion dynamics highlights the extensibility potential of our modelling approach, provided that representative, region-specific covariates and training data are available. This first demonstration of the capacity of machine learning models to cope with the episodic, localised nature of soil erosion on arable land represents a significant advancement in erosion-prediction technology that we anticipate will open new research avenues and lead to new developments in erosion-risk assessments. In particular, implementing erosion nowcasting systems can vastly increase the capacity of stakeholders and decision-makers to react to the erosion hazards that threaten the soil-food nexus.

CRediT authorship contribution statement

Pedro V. G. Batista: Writing - Original Draft, Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Visualisation, Writing - review and edititing. **Markus Möller:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualisation. **Karsten Schmidt:** Writing – review & editing, Methodology. **Timm Waldau:** Writing – review & editing, Resources, Project administration, Methodology. **Kay Seufferheld:** Writing – review & editing, Methodology, Investigation. **Abdelaziz Htitiou:** Writing – review & editing, Validation, Methodology, Investigation. **Burkhard Golla:** Writing – review & editing, Resources, Project administration. **Florian Ebertseder:** Writing – review & editing, Resources, Methodology, Data curation. **Karl Auerswald:** Writing – review & editing, Resources, Methodology, Data curation. **Peter Fiener:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualisation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code availability

The model code is available at https://zenodo.org/records/11072309.

Competing interests statement

Karl Auerswald is a member of Catena's editorial board. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.catena.2025.109080.

Data availability

NowCastR (Original data) (Zenodo)

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